EXPERIMENT – 8

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Design a PlayArena Sports Booking System with AI-based (RAG-inspired) features to provide smart recommendations, player matching, booking insights, and availability prediction.

AIM:

To design and simulate a PlayArena sports booking application that incorporates Retrieval-Augmented Generation (RAG) concepts to provide intelligent recommendations, player matching, booking prediction, and contextual notifications.

Step 1: Designing Login Interface

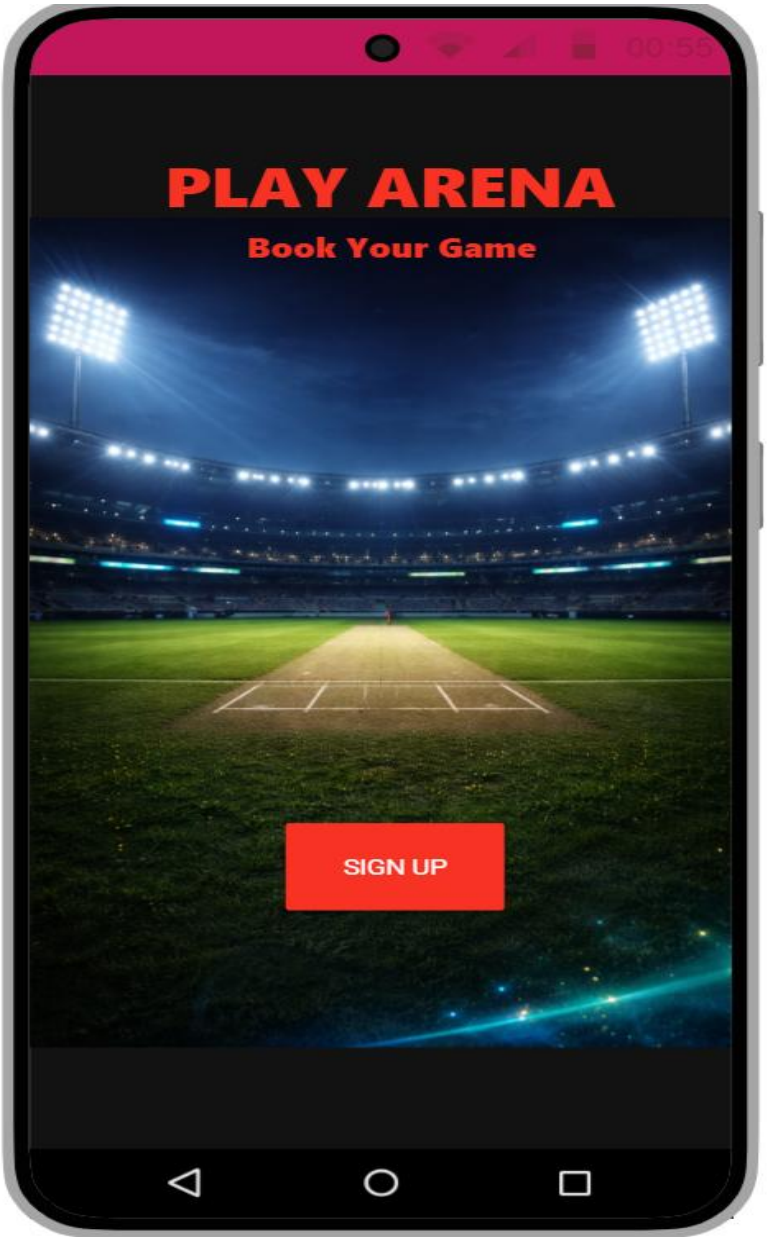
Description:

A login interface is created to allow users to securely access the PlayArena application using email and password.

RAG Concept Applied:

No RAG is applied here since authentication does not require retrieval or generation.

Output:



Step 2: Designing Dashboard (Home Screen)

Description:

The home screen displays sports options like Cricket, Pickleball, and Badminton along with booking access.

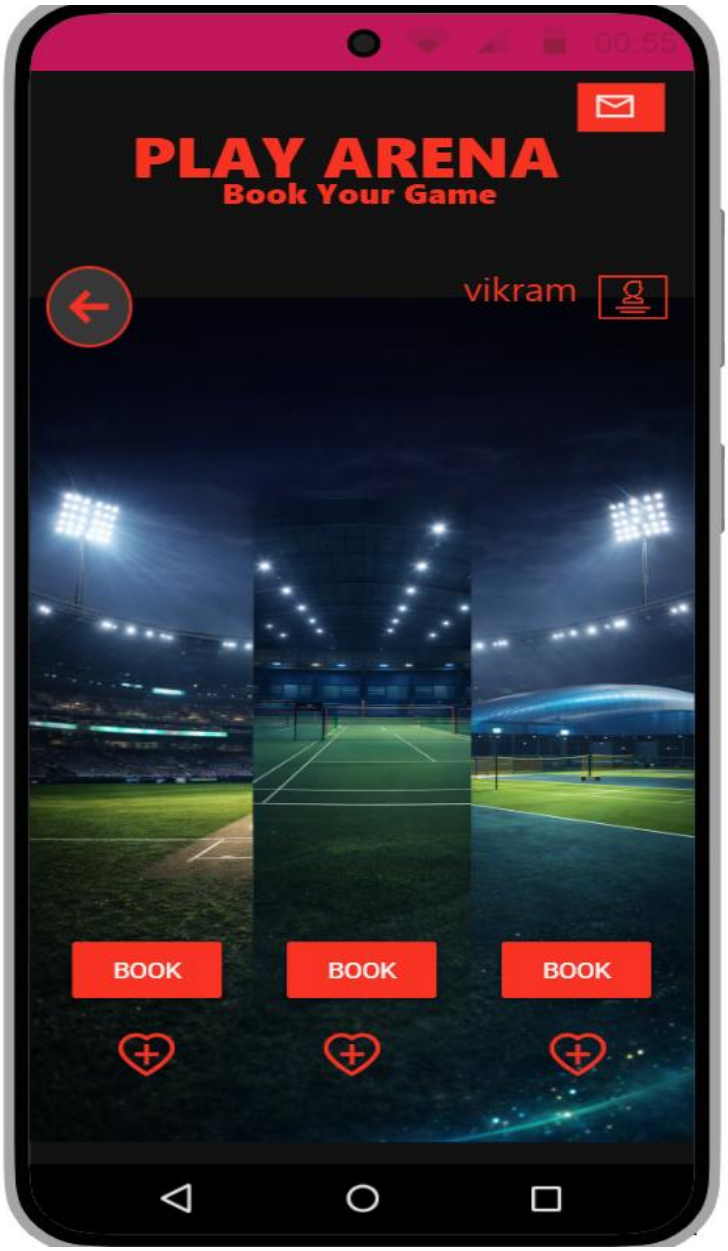
RAG Enhancement:

“AI recommends nearby popular turfs based on recent bookings.”

RAG Logic:

- **Retrieval: Nearby turfs + recent booking trends**
- **Generation: Personalized turf suggestions**

Output:



Step 3: Designing Sport Booking Page

Description:

Users can view available turfs, ratings, and time slots for a selected sport.

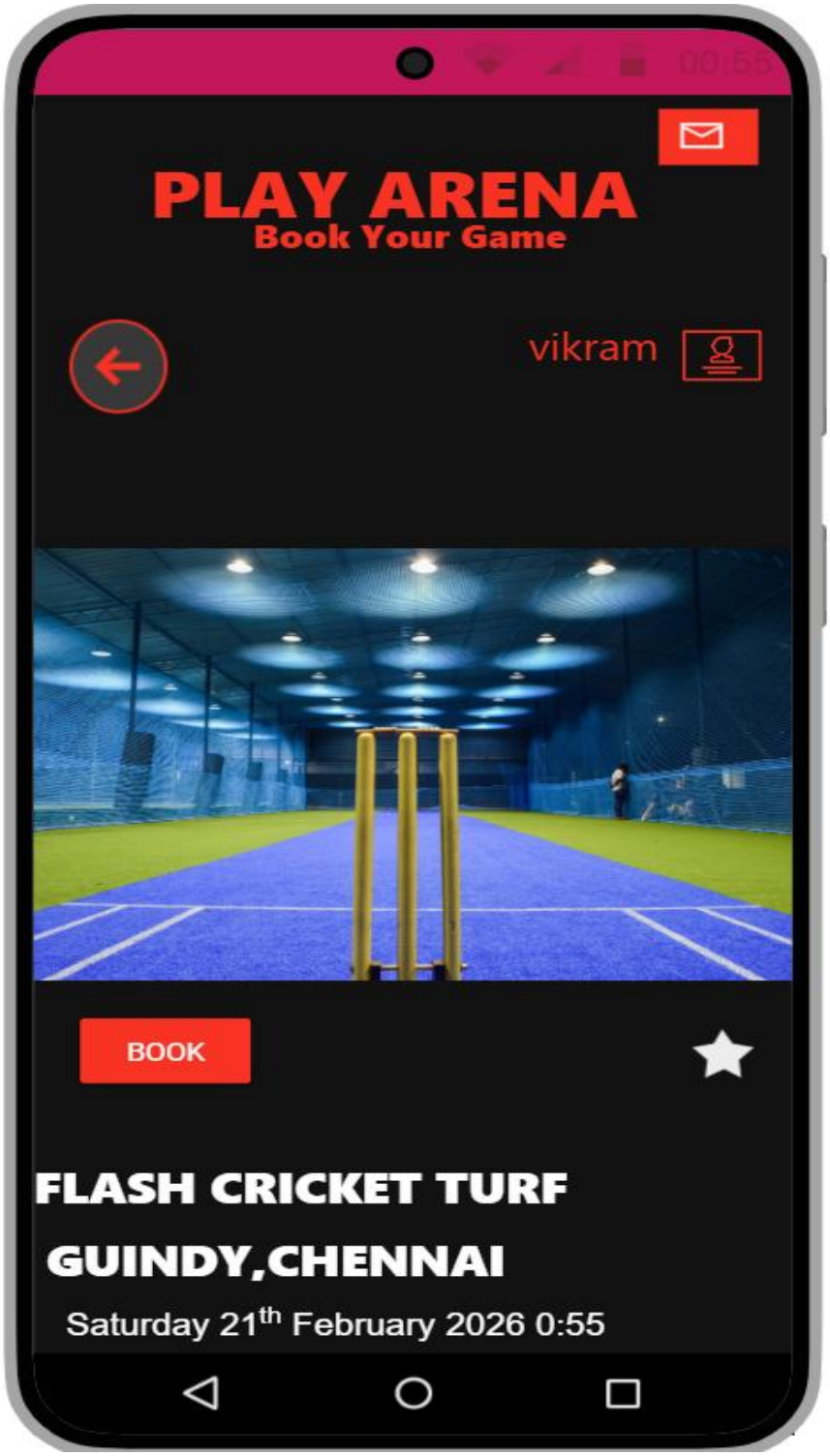
RAG Enhancement:

"AI highlights best time slots based on crowd patterns."

RAG Logic:

- **Retrieval: Past booking data + peak hours**
- **Generation: Optimal time slot suggestions**

Output:


PLAY ARENA
Book Your Game
vikram
BOOK
FLASH CRICKET TURF
GUINDY,CHENNAI
Saturday 21th February 2026 0:55

Step 4: Designing Booking Confirmation Page

Description:

Users confirm their selected turf and time slot.

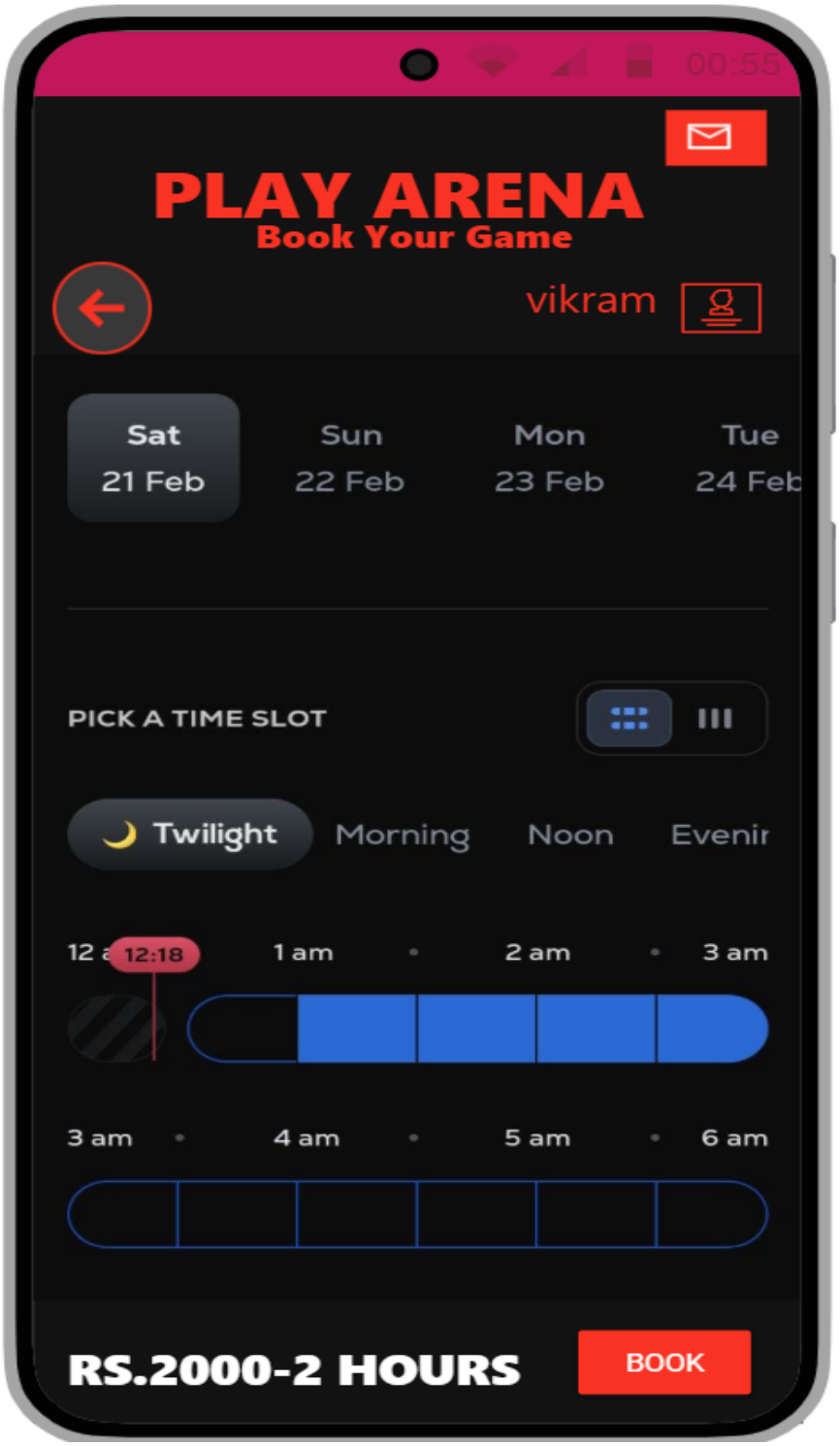
RAG Enhancement:

“AI predicts high demand and suggests alternative slots.”

RAG Logic:

- **Retrieval: Real-time + historical demand data**
- **Generation: Predictive recommendations**

Output:



Step 5: Designing Final Confirmation Screen

Description:

Displays booking success and details.

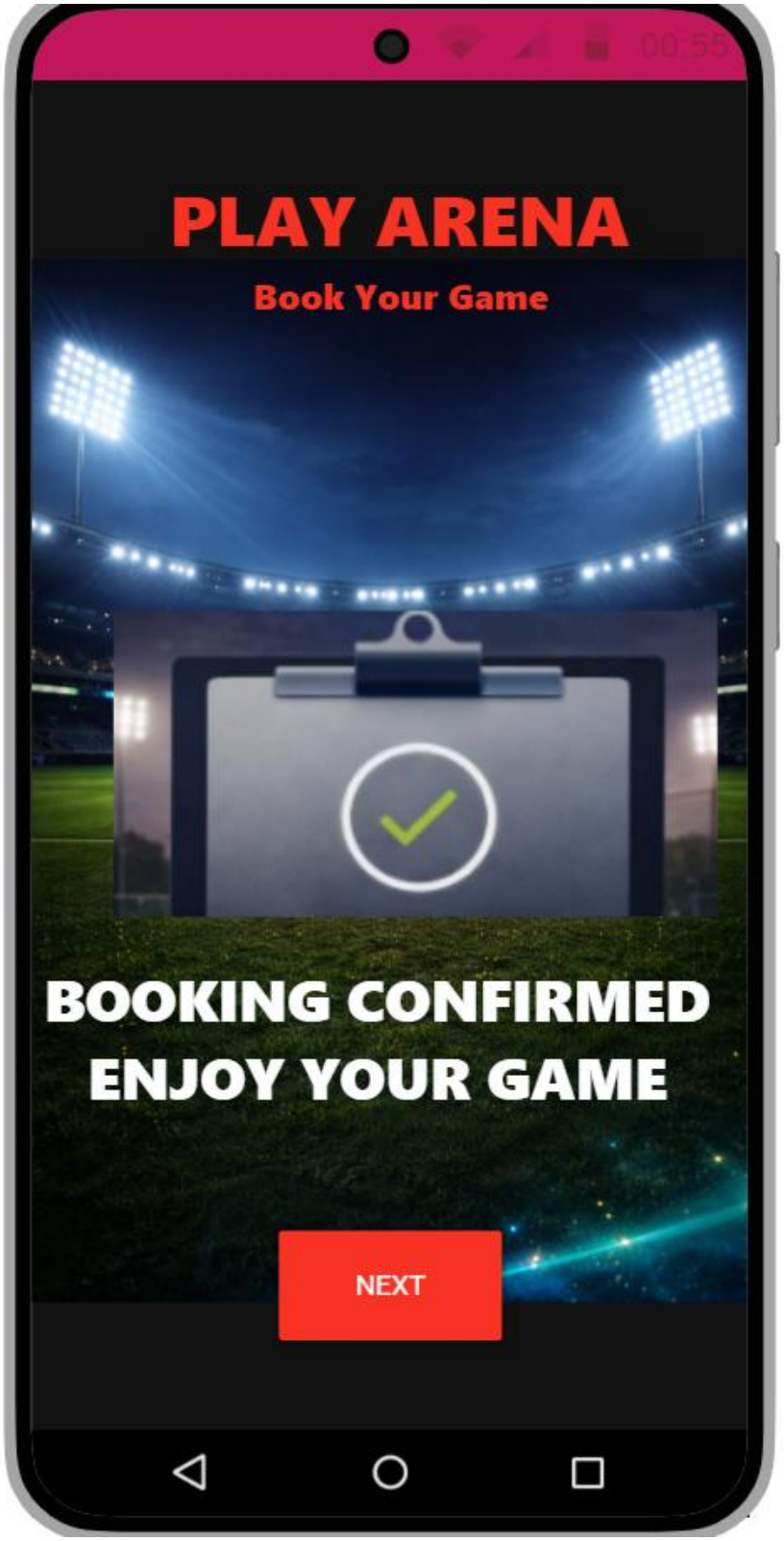
RAG Enhancement:

“AI found nearby players with similar bookings.”

RAG Logic:

- **Retrieval: Nearby users + similar bookings**
- **Generation: Player matching suggestions**

Output:


PLAY ARENA
Book Your Game
BOOKING CONFIRMED
ENJOY YOUR GAME
NEXT

Step 6: Designing Notification System

Description:

A notification panel provides booking updates and suggestions.

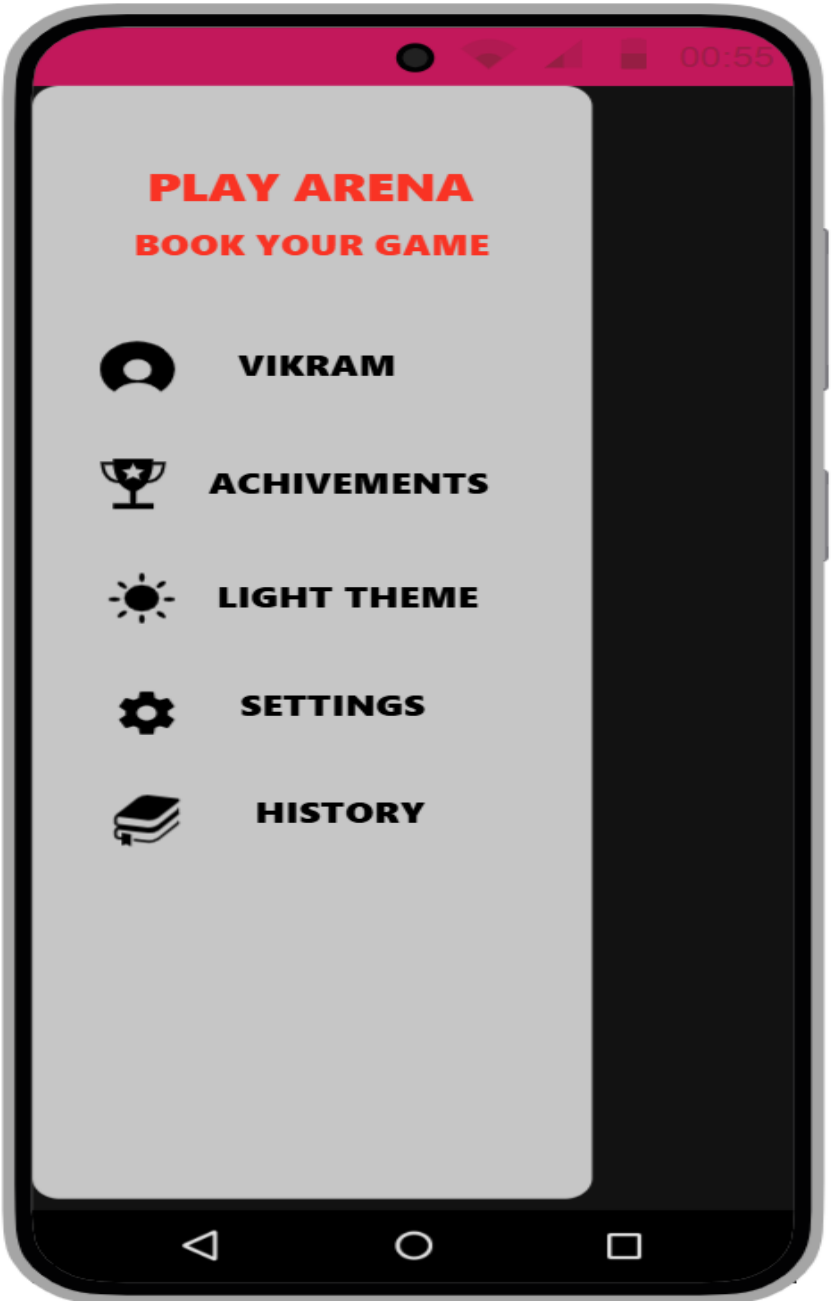
RAG Enhancement:

“AI matched your booking with similar games in your area.”

RAG Logic:

- **Retrieval: Local activity data**
- **Generation: Contextual notifications**

Output:



PROTOTYPE LINK:

https://pr.to/PM1LTS/

RESULT:

The PlayArena system was successfully designed with RAG-inspired features such as smart recommendations, booking prediction, and player matching.